

St George's Field

Pavilion

initial ideas



Concept: An organic curved pavilions inspired by natural forms, creating a smooth flow from the ground up. The design forms a sheltered space, while the glass openings maintains a strong connection outdoors

Key Features: Smooth curved form- unique visual identity Large glass openings.

Materials: Curved laminated timber-flexible, allows curved forms.

Glass panels- brings in natural light and maintains connection to surroundings

Pros: Unique design and shape that resonates with surroundings

Cons: Not much privacy- glass openings make everything inside visible, Complex construction

Complexity level: Complex, curved structure requires skilled

Materials: Panels- structural skin and wall. Glulam beams- Reinforces cantilever. Glass panels- allows light to enter

Key Features: Excessive cantilevered volumes Sculptural but structural composition

Pros: Natural lighting Cheap materials

Cons: Structurally complex Needs precise engineering and detailing

Complexity level: Complex- project specific engineering, complex load

paths, need to be very precise



Concept: A continuous curved shell with a large circular incision split in the middle forming the entrance, the design mimics a cave-like space embedded within the landscape



Material: Brick-Masonry- Structurally durable Key Features: Incision creating the entrance Heavy grounded- blends into the ground space rather than differentiating itself from it Semi-enclosed interior environment

Pros: Strong durable structural system Natural material integration

Cons: Limited natural light, Project specific engineering

Concept: An organic pavilion inspired by natural cellular forms creating a sculptural and immersive space while providing a semi-enclosed space

Materials: Concrete (composite shell)- strong and is easily malleable. Perforated surface- tiny pores which allows air and light to get through.

Key Features: Perforations creates an interesting light pattern. Organic form blends in with natural concepts

Pros: Unique design, mechanically efficient- no need for glass panels for light due to perforation

Cons: Expensive to build

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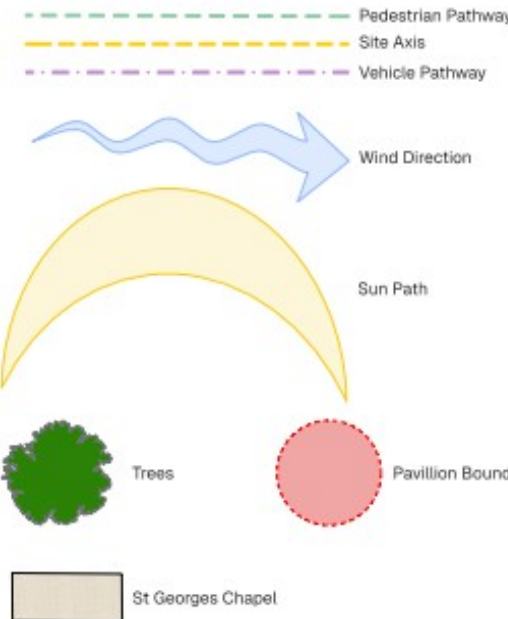
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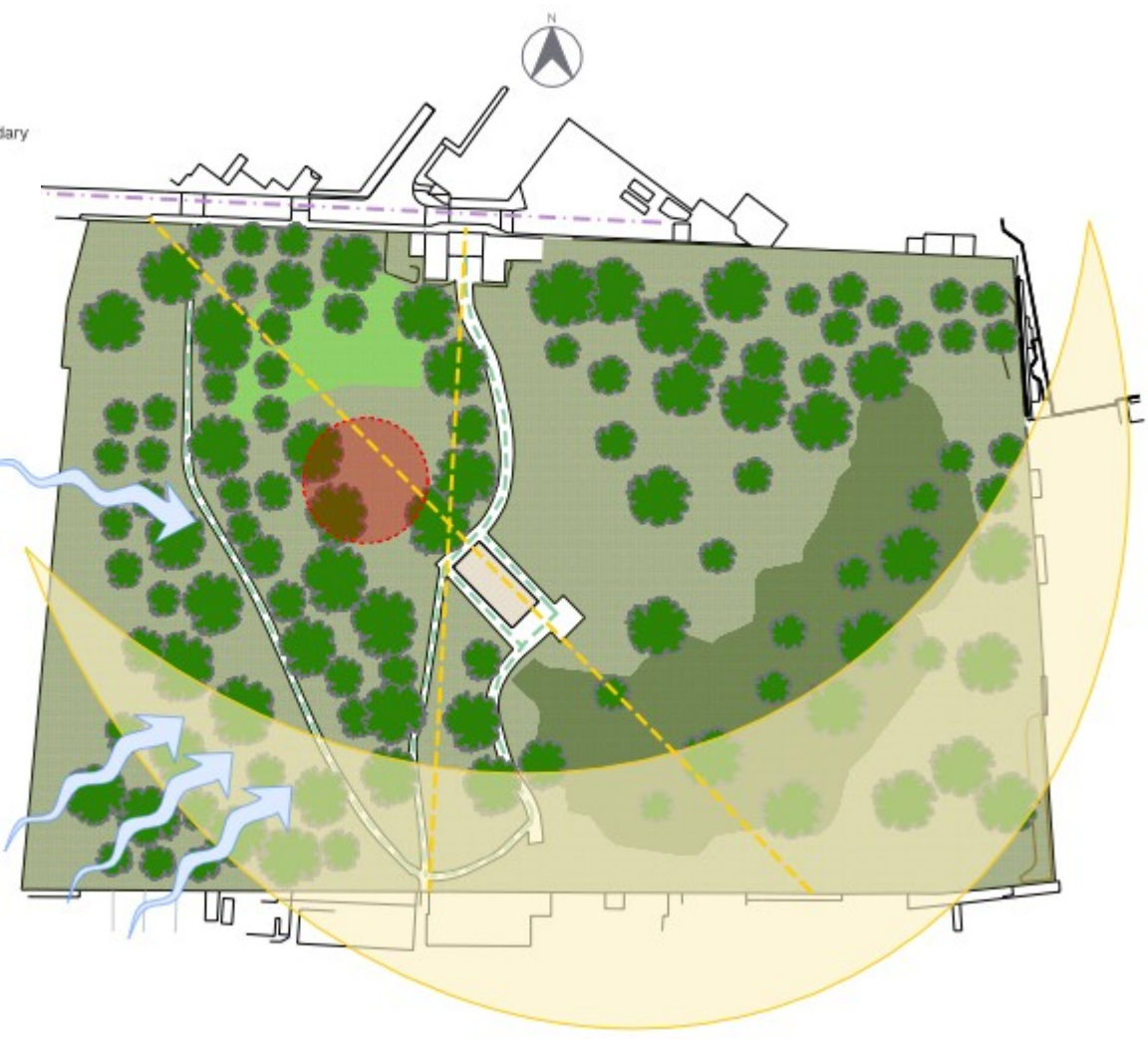
Cons: Expensive to build

Precedent research

Material	Cost	Advantages	Disadvantages
Reinforced concrete	150 – 250 £/m ³	Strong, durable, low maintenance	High embodied carbon, heavy
Glulam timber	1000 £/m ³	Natural look, sustainable, warm	Requires maintenance, fire risk
Structural steel	50 – 100 £/m ³	High strength, long life span	Corrosion risk, costly finish (for large projects)
Corten steel	150 £/m ³	Self-weathering, unique aesthetic	Heavy, stains surroundings
Polyester canvas	2.50 – 10 £/m ²	Moisture and Mold resistant	Plastic look and feel, static electricity
glass	35 –>100 £/m ²	Seamless connection between interior and exterior	Greenhouse effect, poor insulation, can be fragile

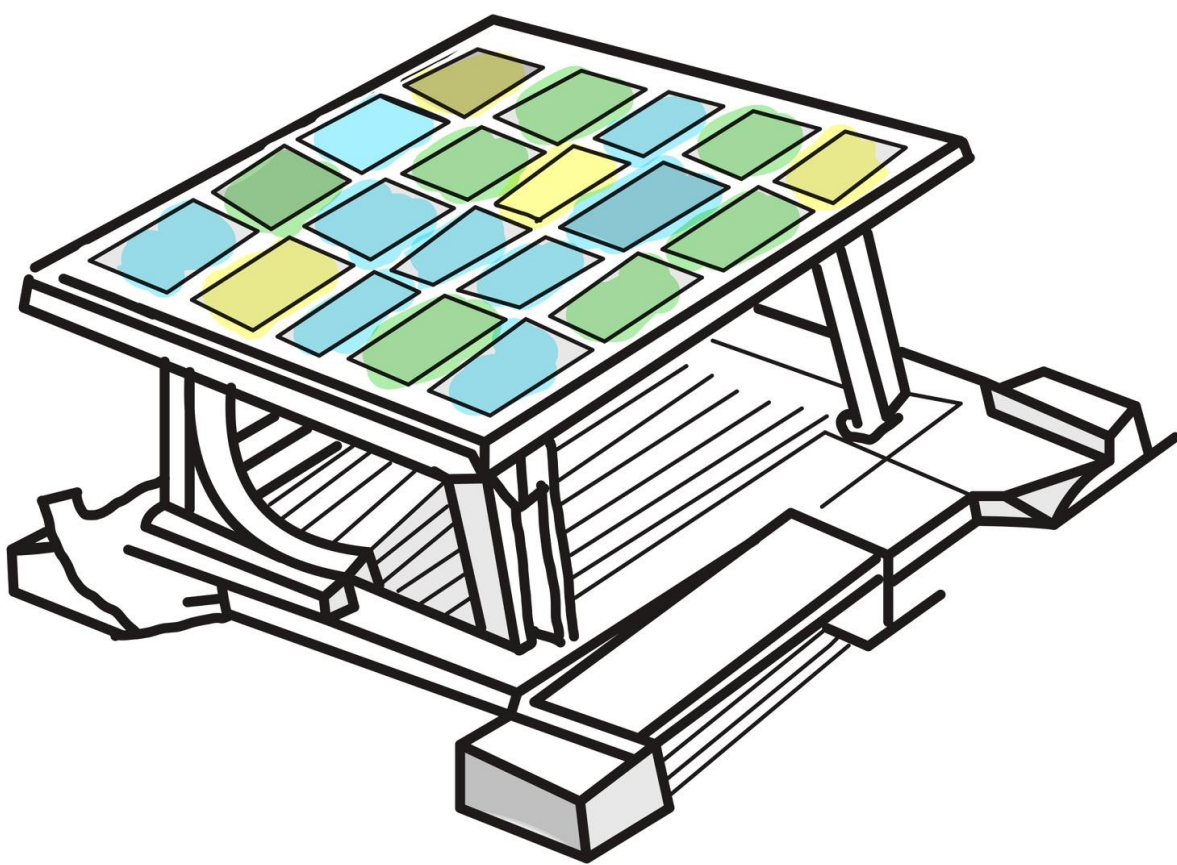


Site Analysis



Initial Design Ideas

Design idea 1

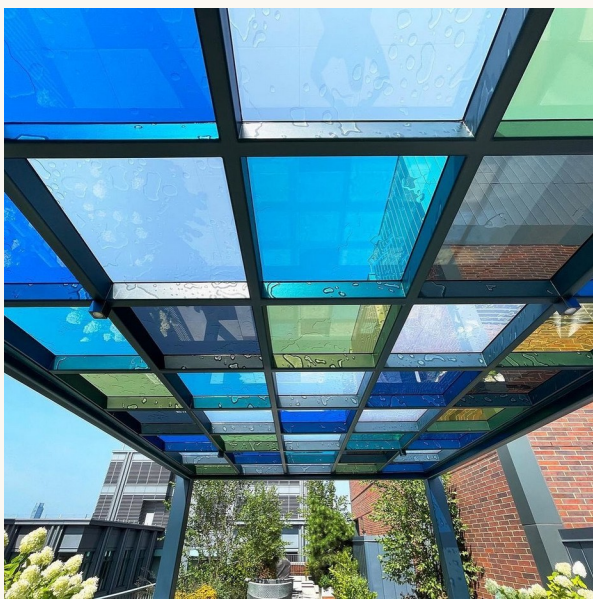


Concept:

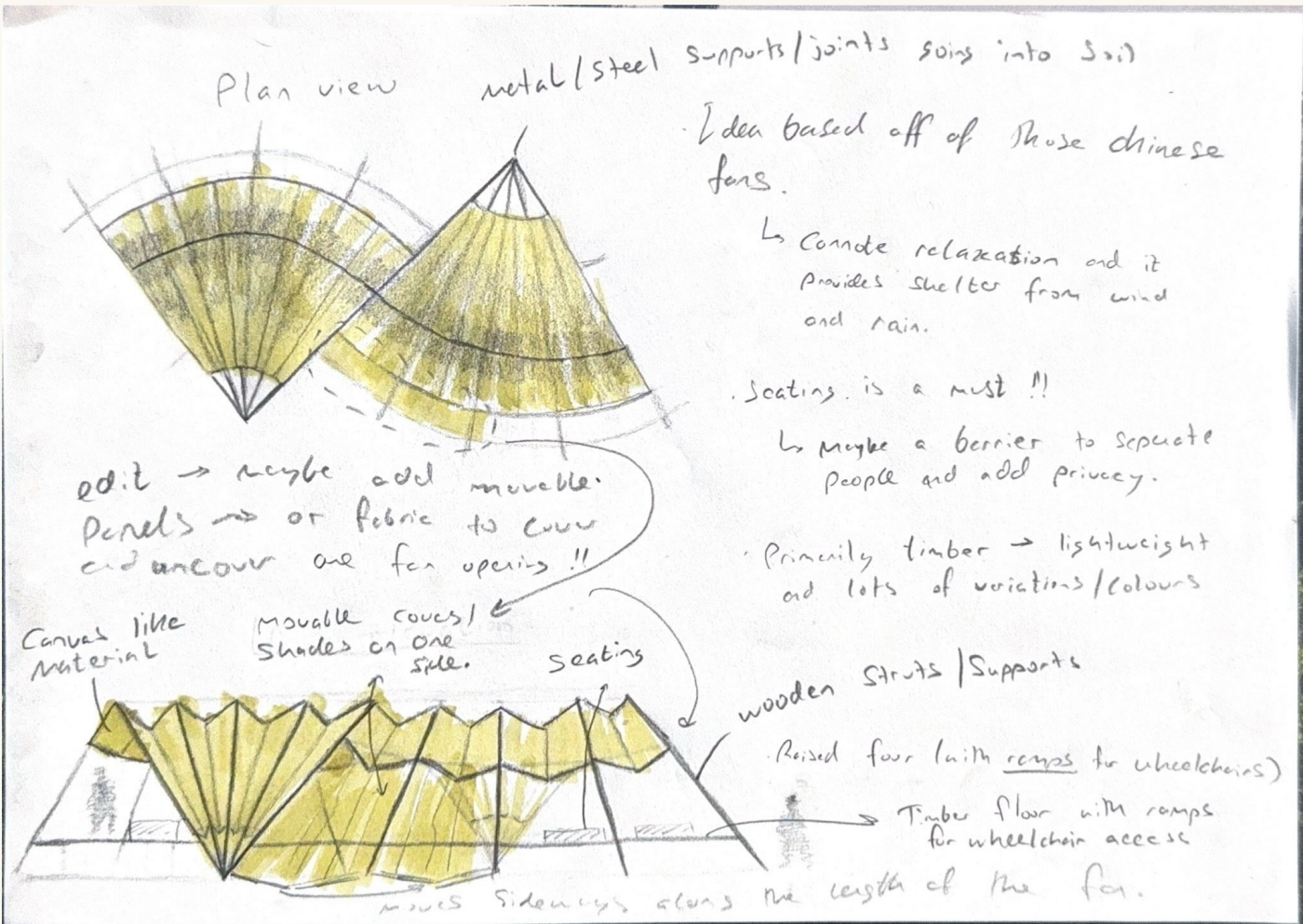
To provide shelter as well as a source of energy through solar power. The simple, cubic shape gives a sense of security and therefore, could increase comfort.

Key features:

- Monocrystalline solar panels
- Tempered glass roof to allow for sunlight
- Recycled steel



Design idea 2



Concept:

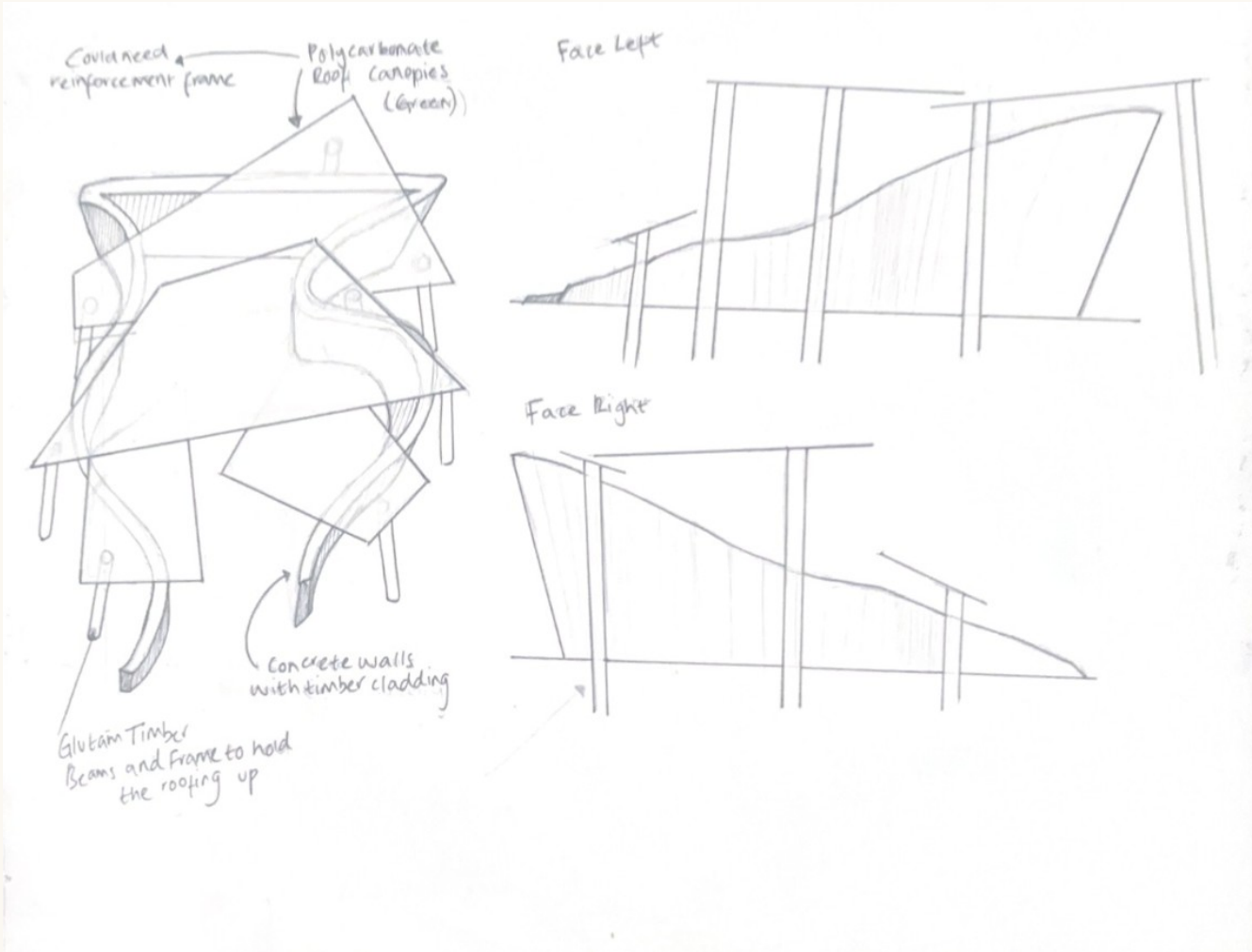
Inspired by the handheld fans people may use to cool themselves down, as well as those bamboo panels used as a way to separate two areas for privacy.

Key features:

- Natural colours
- lightweight roof/structure
- Adjustable shade on one side



Design idea 3



Concept:

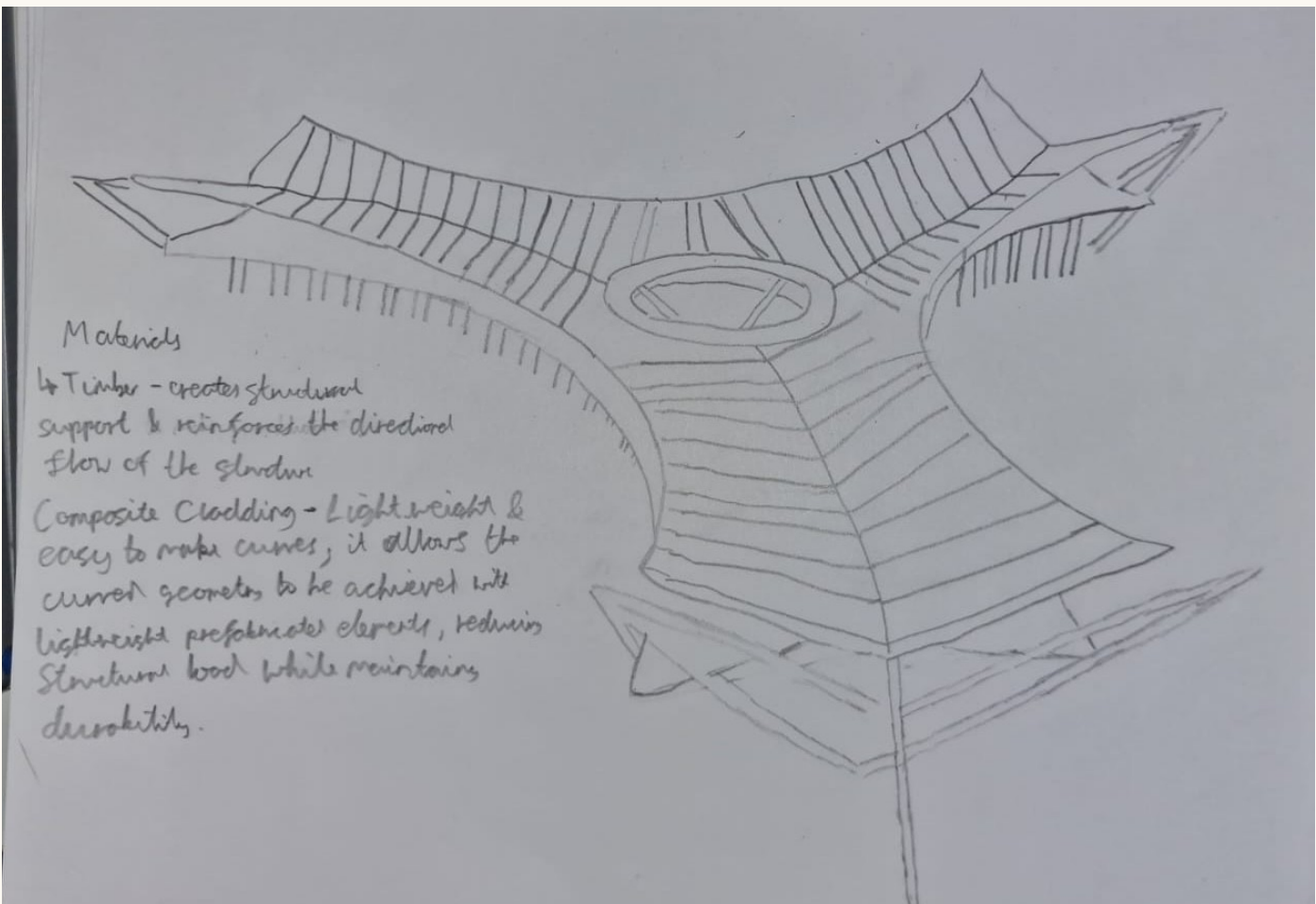
Inspired by the shape of a high-heels with curved walls to mimic the fluidity of water

Key features:

- Poly-carbonate roof canopies
- Timber beams and frames
- Open walkway + space



Design idea 4



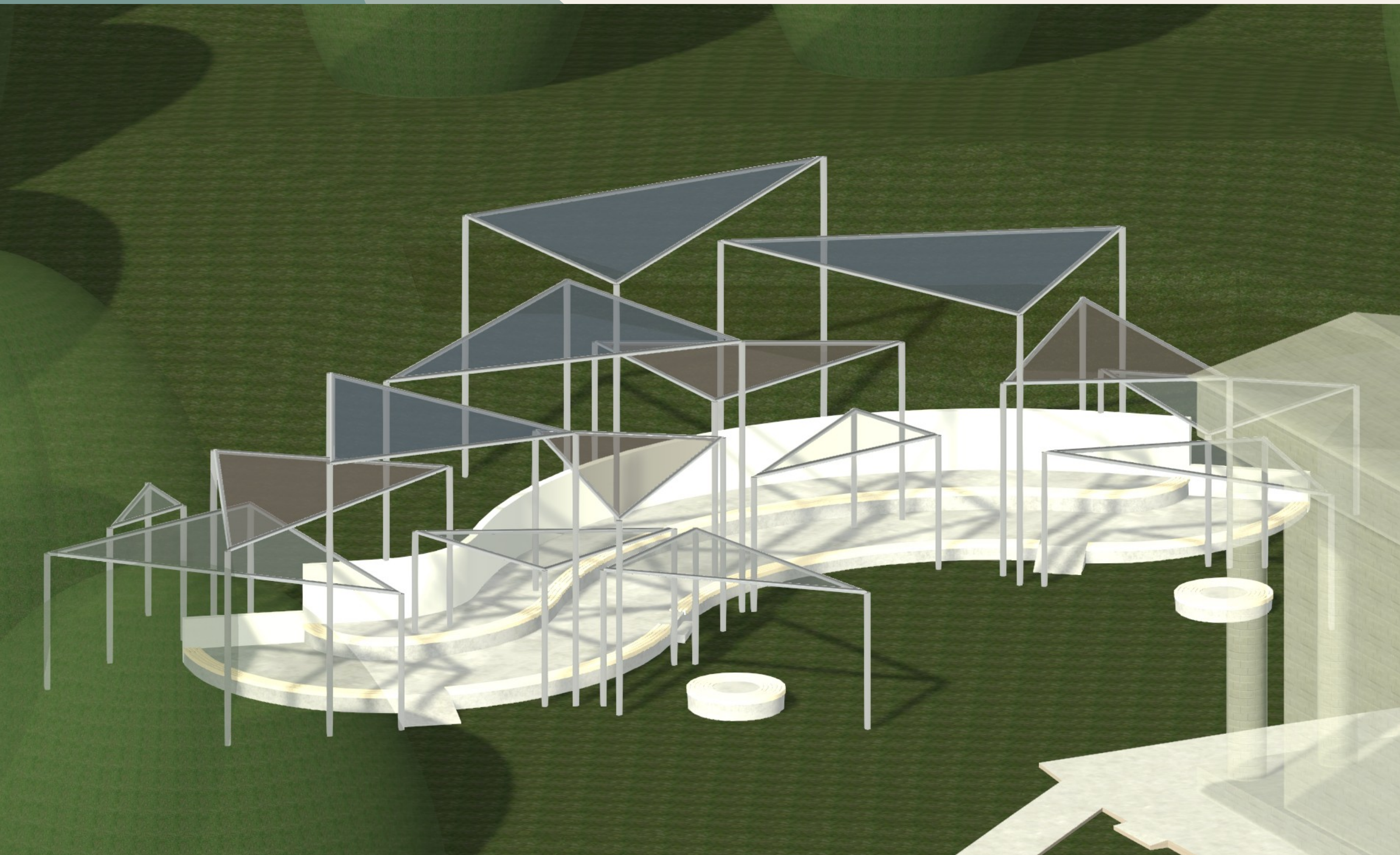
Concept:

To provide a relatively large area of relaxation and comfort through the use of timber as the main material and cladding to provide a sense of shelter and cover.

Key features:

- Timber structure
- Composite cladding
- Large, curved roof

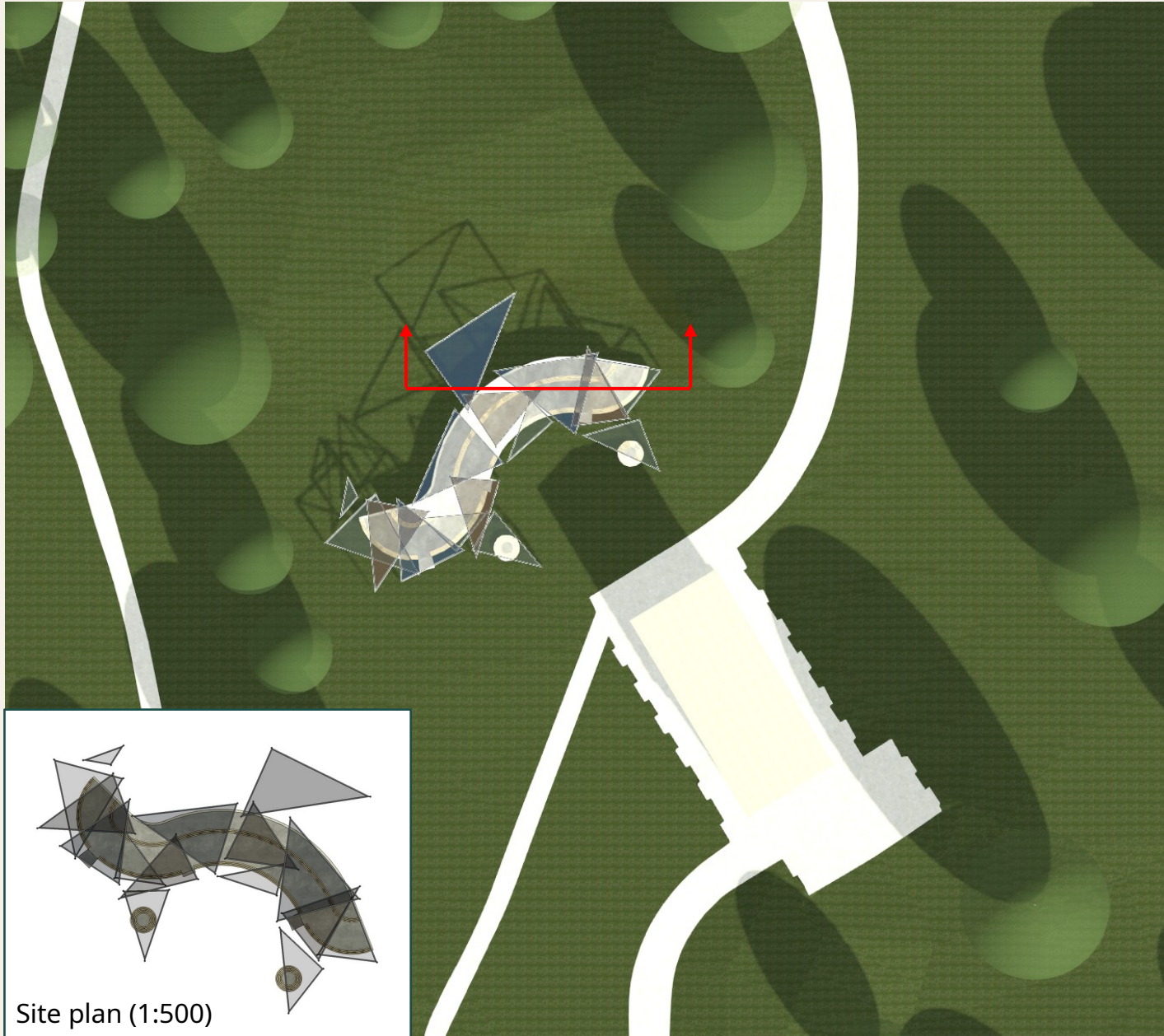




3D Orthographic
(1:100)

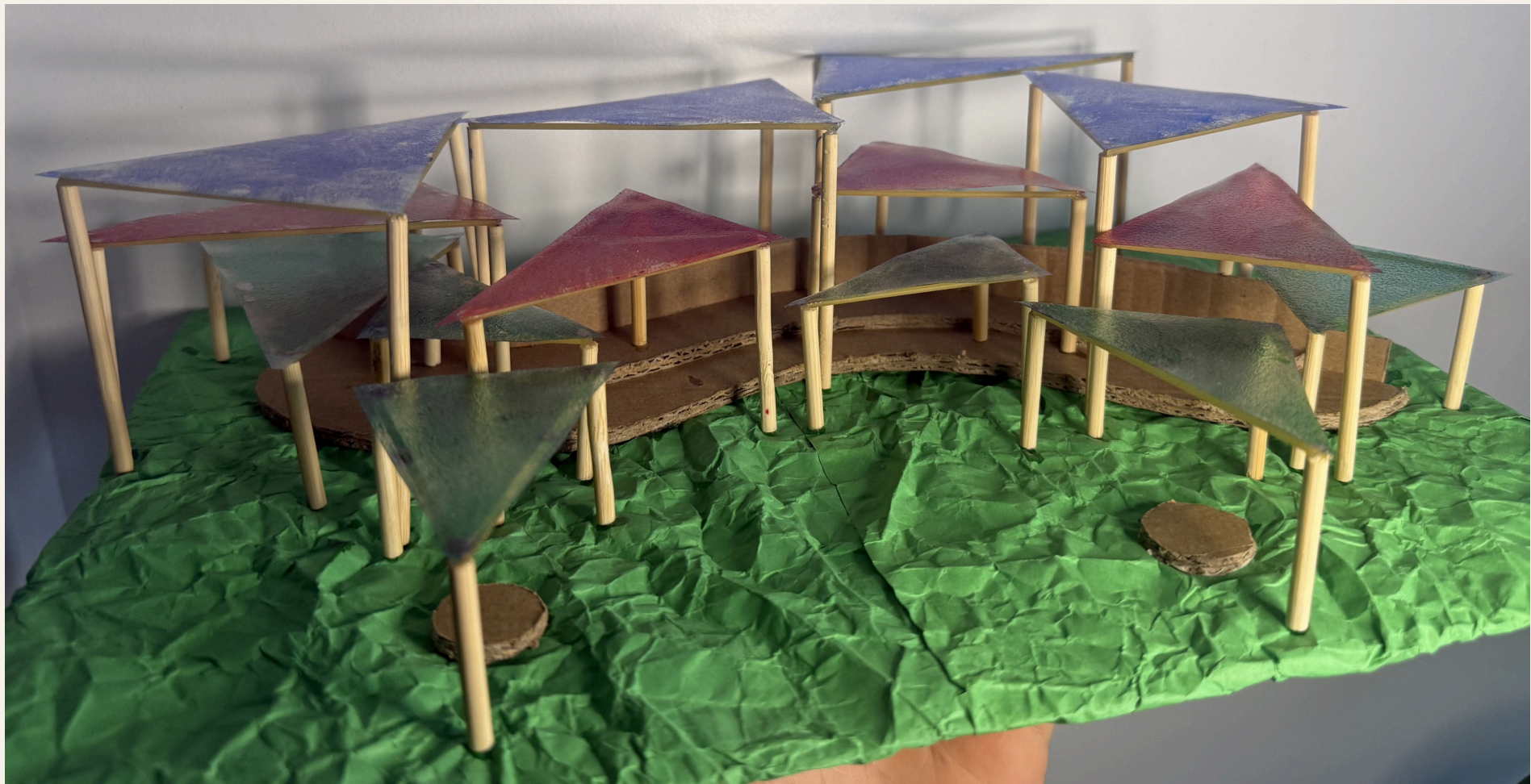
Design concept

Our idea was to include context behind St George and his accomplishments while still making the pavilion functional as a place of relaxation/study sheltered from the elements but within them at the same time. The tall roof is there to make the space seem more open; and the shape of the base to imitate that of a couch. However, the 45 columns are there to provide a sense of safety and support, through their tall yet solid structure. When looked in plan view the panels' position and orientation resemble that of a dragon. This relates to the tale of how St George defeated a dragon to save a princess. The height of the roof panels further emphasizes this through their tall height (maximum 7000mm).



Site plan (1:500)

Physical model (1:100 scale)

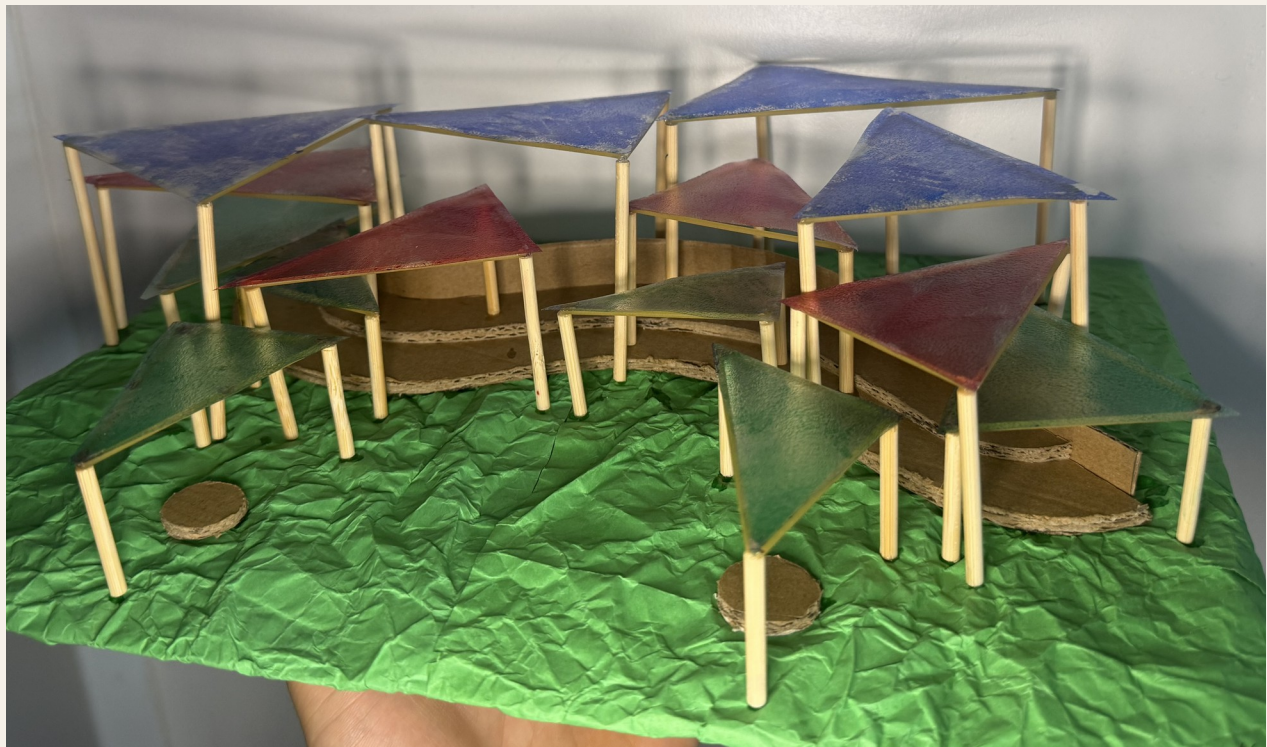


First perspective (1:100)

Model information

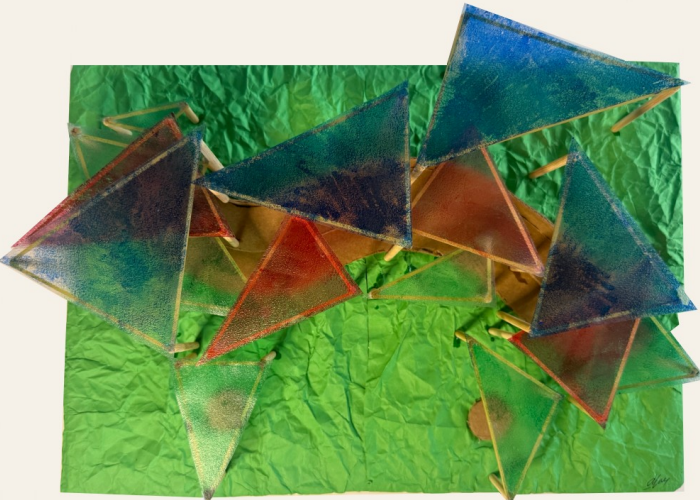
The model was constructed using various resources:

Cardboard;
Coloured paper;
Bamboo sticks;
Spaggetti;
Plastic wallets;
Sharpie felt-tips;
Super-glue + activator and;
Clear tape.

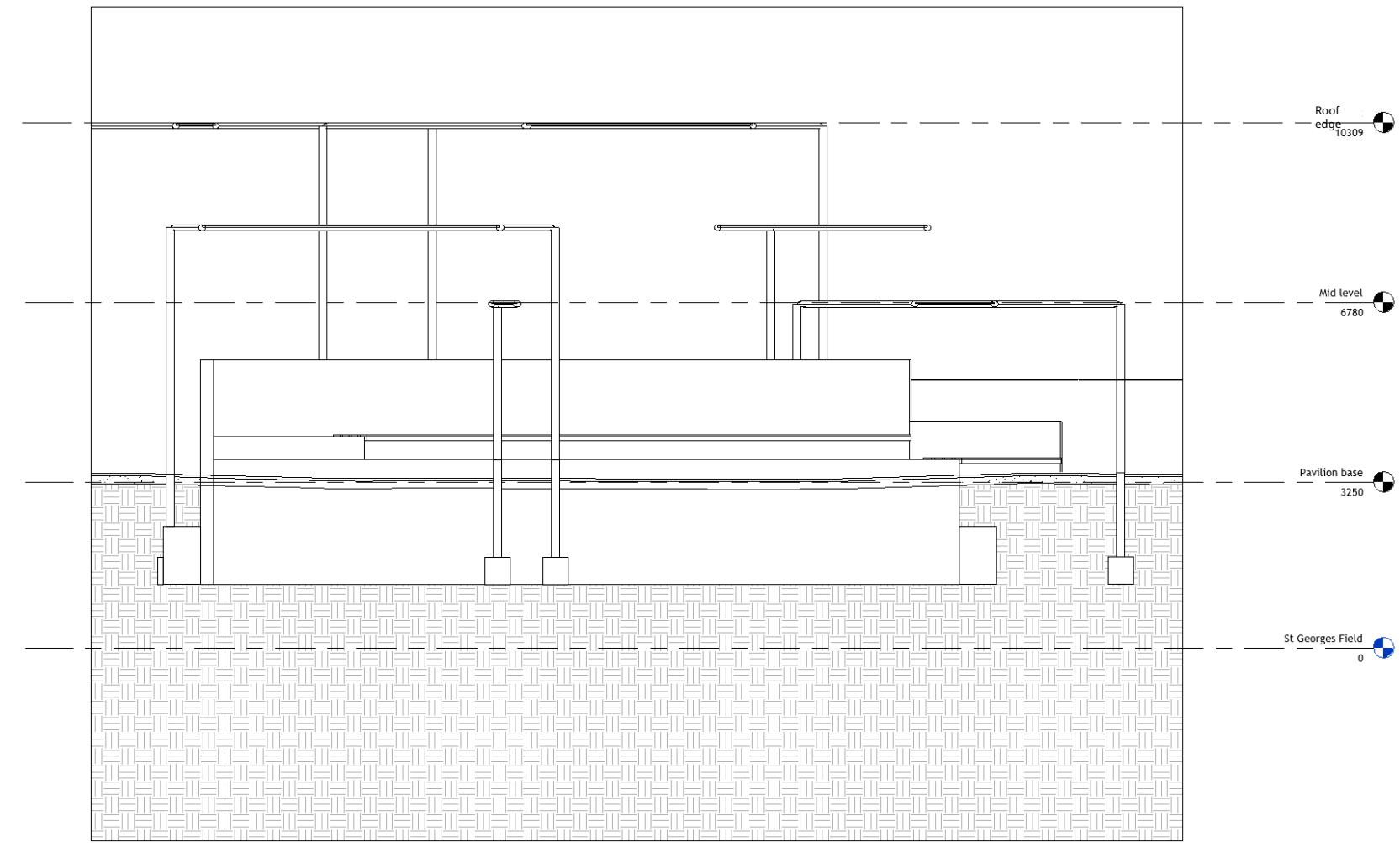


Second perspective (1:100)

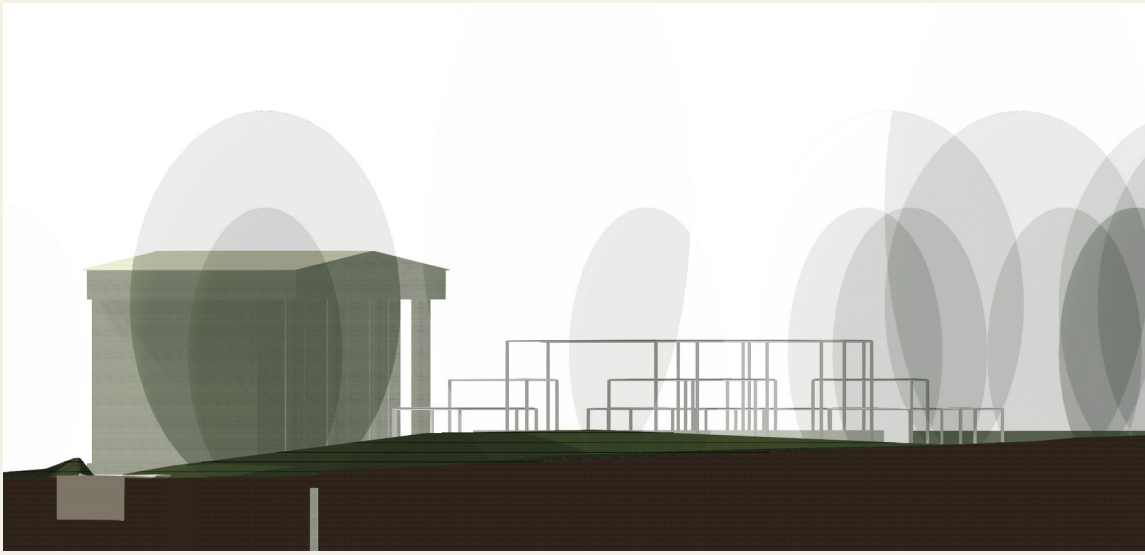
It is important to note that the model scale is only as estimate of 1:100 as it was challenging to measure and cut exact dimensions of the curved seating. Furthermore, the steel columns have a diameter of 180mm (as seen in the glass panel construction detail), this would mean that the 1:100 scale dimension for the model should be 1.8mm which is extremely thin, and therefore, the aluminium beams would be even thinner. This proved extremely difficult to work with so the scale for the columns and the beams are not accurate. However, the seating, and the size of the glass panels are to scale.



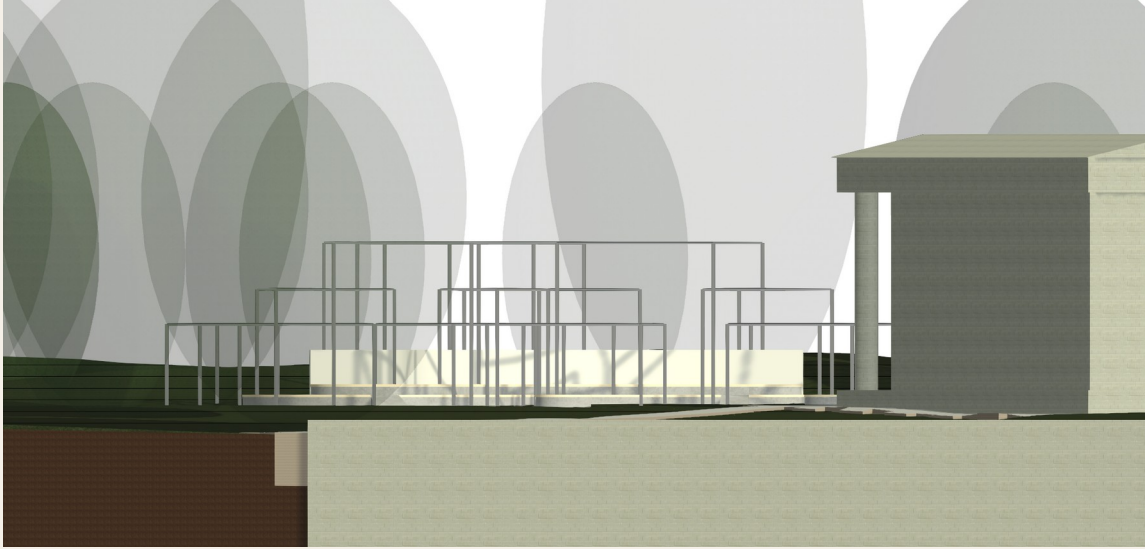
Plan view (1:200)



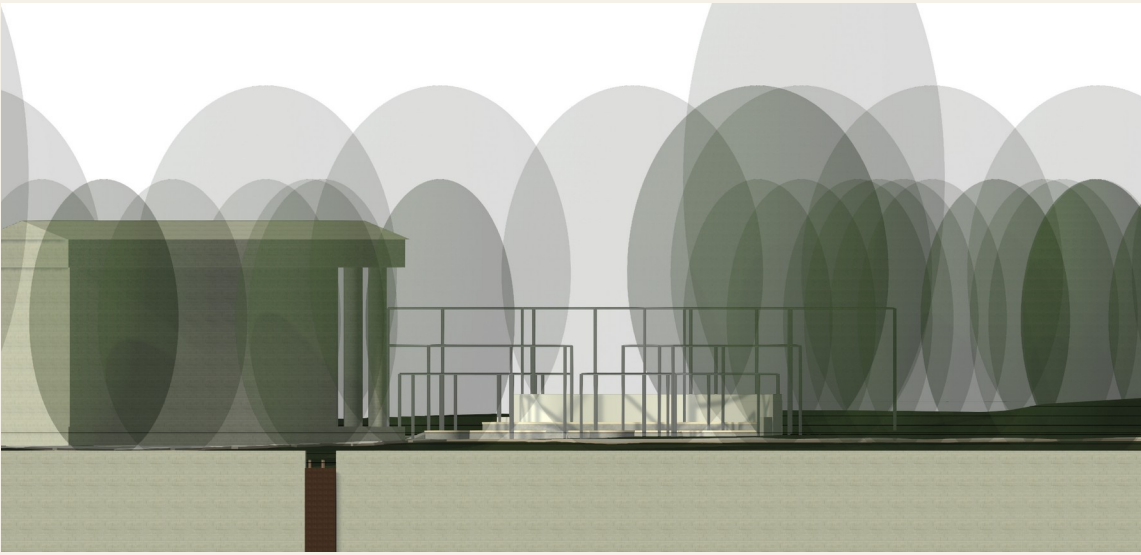
Section drawing (1:10) → section location shown in site plan



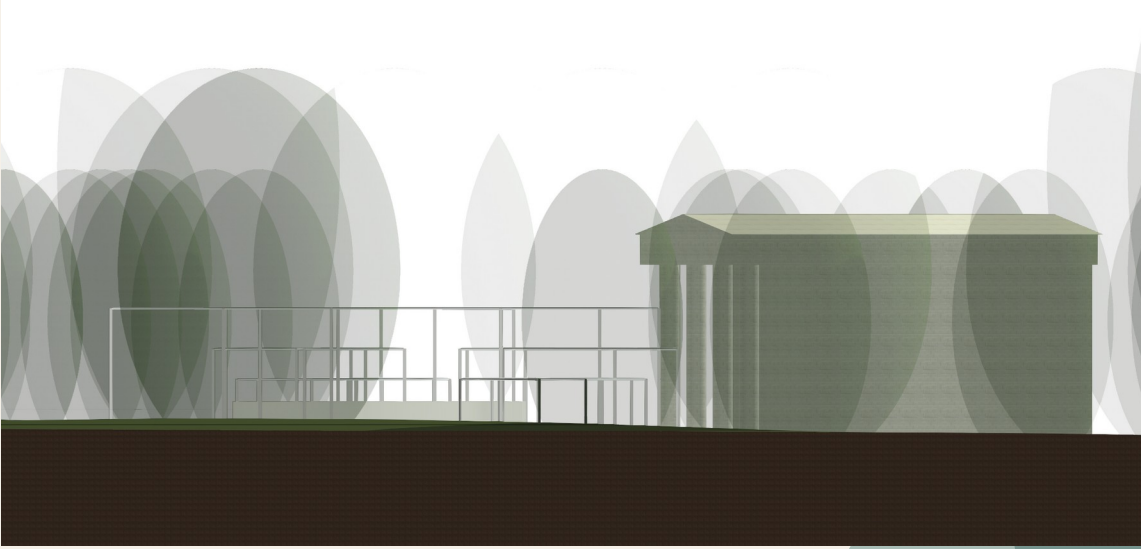
North elevation (1:200)



South elevation (1:200)

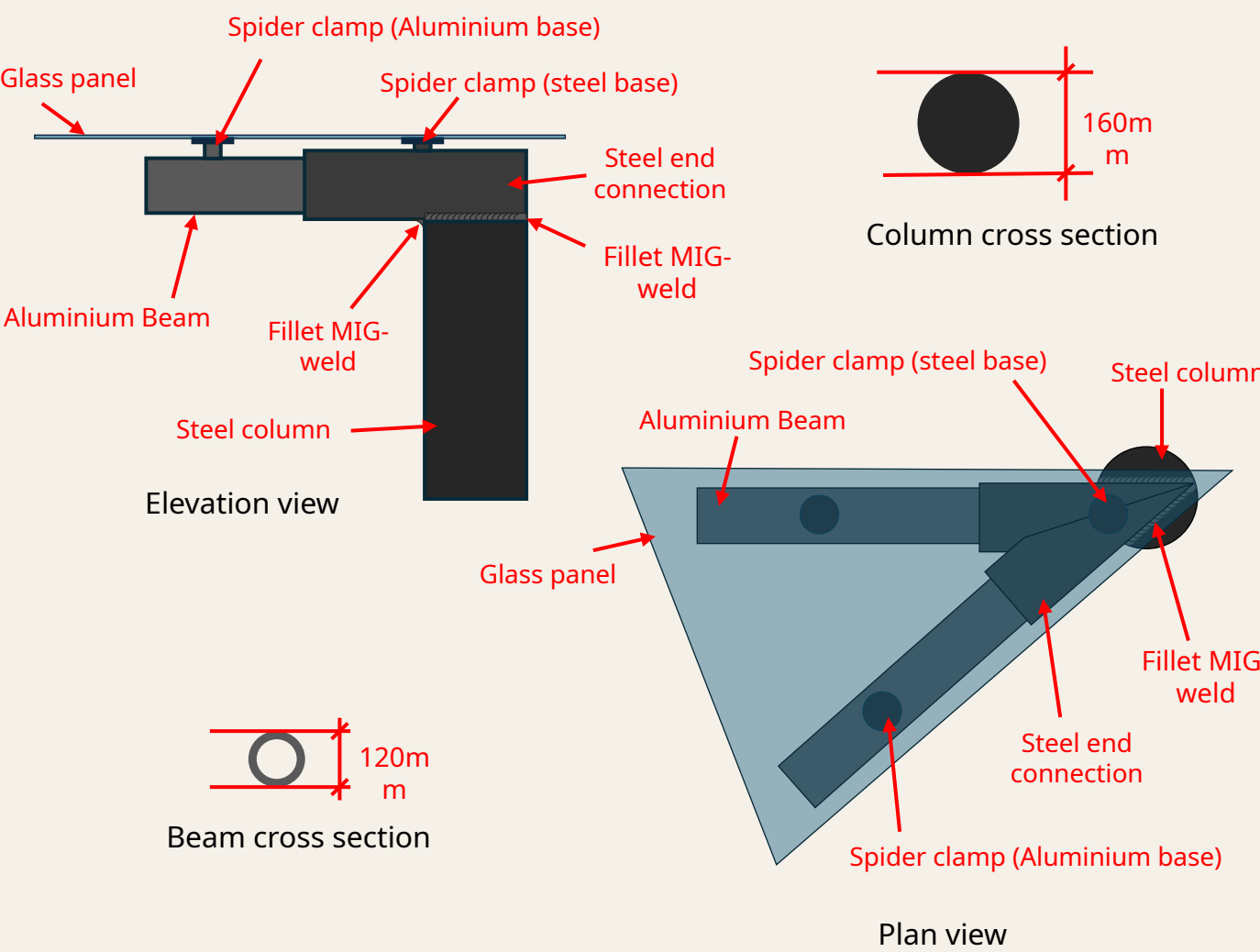


East elevation (1:200)



West elevation (1:200)

Glass panel construction detail (1:5)



Canopy and enclosure

Glass panels are scattered over the pavilion in specific orientation and position as to for the shape of a dragon. The panels are tinted by a special material that absorbs UV radiation and therefore prevents magnification of most UV radiation, keeping the underside of the canopy cool.

Floor

Features two levels, a lower, wider area and an upper, more narrow area; both constructed out of concrete. The floors are stacked and have been constructed with wooden panels that raise the seat height from 450mm to a more comfortable 480mm. Ramps are also constructed for those who may need them.

Form and structure

In plan view, the shape of the pavilion takes the shape of a dragon (in context of the tale of St George), providing a sense of connectivity to St George and the site. The seating of the pavilion is relatively low, with its second level concrete base only standing at 900mm and its first at 450mm from the base of the structure. The slender columns may seem unstable and wea to buckling, however, they are constructed completely out of steel and are solid; they protrude 2000mm into the surface from the base of the pavilion. The use of triangles as the roof panel shape also provides security through their inherited high-strength design.

construction method

1. Firstly, the site must be prepared for excavation, this requires soil investigations and removal of plant/vegetation on the surface. Then the excavation must be dug (with appropriate shielding and safety procedures in place). Including excavating the holes for the columns and their foundations.
2. The foundation rebar and concrete must be placed and let to settle and dry. Then the construction of the base of the pavilion may be started. To reduce the embodies carbon, the base will be pre-cast and then transported to site; steel connections to be used to connect the blocks and filler to fill in the gaps.

Material cost estimation

Material	Cost per unit ...	Total area (m ²)	Total volume (m ³)	Total cost (£)
Reinforced Concrete	200 £/m ³		464.087	92817.4
Steel	75 £/m ³		4373.1	327982.5
Aluminium	5994 £/m ³		1.195	7162.83
Spruce wood	800 £/m ³		7.673	6138.53
Glass	100 £/m ²	334.43		3344.3
Total cost				437445.56

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